



VAULTX CYBER TECH

BEYOND TODAY · BEFORE TOMORROW

SUMMER INTERNSHIP PROGRAM 2026

WEEK

04

RAG — AUTOMATION OVER YOUR OWN DOCS

CHUNKING / EMBEDDINGS / VECTOR DB / GROUNDED ANSWERS / CITATIONS

6

TASKS

6

DELIVERABLES

INTERMEDIATE

LEVEL

ONSITE / LAB

MODE

vaultxcybertech.com

info@vaultxcybertech.com · +92 306 2000928



WEEK FOCUS

Objective	Build a retrieval-augmented system so automations can answer from real, private documents — the core of most business AI — with citations and honest 'no answer' handling.
Skills Targeted	Chunking Embeddings Vector DB Retrieval Grounded generation Citations
Tools & Stack	Anthropic / OpenAI embeddings Chroma / FAISS / pgvector LangChain or plain Python a real document set

TASK LIST

01 Ingest a real corpus

- Take a real document set (company docs, PDFs, a knowledge base).
- Clean and chunk it sensibly (size + overlap) and explain your chunking choices.

02 Build the index

- Embed the chunks and store them in a vector DB (Chroma / FAISS / pgvector).
- Persist the index so it survives a restart.

03 Retrieval + grounded answers

- Given a question, retrieve top-k chunks and answer ONLY from them, with citations to the source chunks.
- Refuse when the answer isn't in the documents.

04 Evaluate honestly

- Build 15–20 QA pairs; measure how often answers are correct and grounded.
- Log failures and improve retrieval (k, chunking, reranking).

05 Wire RAG into an automation

- Expose your RAG as an endpoint / tool and call it from n8n / Make (e.g. Slack question → RAG answer with sources).

06 Add guardrails

- Handle 'no answer found', add a hallucination check, and always show sources so users can verify.



DELIVERABLES

#	DELIVERABLE	FORMAT
1	Ingestion + chunking script	PY
2	Persisted vector index	FILES
3	RAG query app with citations	PY
4	Grounding evaluation report	PDF
5	RAG connected to a live workflow	JSON / PNG
6	Guardrails writeup + demo	PDF + MP4

SUBMISSION PROTOCOL

Channel	Upload to the VaultX LMS under Internship > AI Track > Week 04.
File Naming	VX_AI_W04_FirstName_LastName.pdf plus one ZIP with all supporting files, exports and screenshots.
Deadline	Sunday, 11:59 PM (PKT) of the assigned week.
Evidence	Every screenshot must show your machine clock and the relevant page/scope.
Walkthrough	Record a short screen video (3–5 min) explaining what you did, what you found, and how you stayed safe.